

Regularized Alpha Maps Through Convex Portfolio Optimization

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Abstract

We fit linear transformations of raw alphas that we use in convex portfolio optimization to maximize the out-of-sample Sharpe ratio using gradient descent. Each horizon has its own map, regularized towards the conditional expectation (CE) assuming joint Gaussianity, or towards its linear projection for historical returns. We compare six synthetic problems and three historical-stock problems, using 10 seeds for each comparison and including a scaled-CE baseline. Hyperparameters are selected on tuning data, with performance evaluated on a separate test set. These comparisons do not establish general superiority of the fitted linear transformation over CE. We test whether each fitted map can be interpreted as a regression on a linear transformation of that horizon’s return vector alone.

1 Motivation

For excess returns r , write $m(a) = E[r \mid a]$ and $V(a) = \text{Cov}(r \mid a) \succ 0$. Using $m(a)$ as the mean input to unconstrained Markowitz optimization gives $w(a) = \gamma^{-1}V(a)^{-1}m(a)$, the direction maximizing conditional Sharpe. Positive scaling does not affect conditional Sharpe. Maximizing unconditional Sharpe over unrestricted state-dependent portfolios instead gives [1]

$$w^*(a) \propto E[rr^\top \mid a]^{-1}m(a) = \frac{V(a)^{-1}m(a)}{1 + m(a)^\top V(a)^{-1}m(a)},$$

with one positive proportionality constant across states. Thus the appropriate Markowitz mean input is proportional to $m(a)/(1 + m(a)^\top V(a)^{-1}m(a))$, which generally is nonlinear in alphas. This distinction motivates learning the input map for the portfolio criterion. The learned maps are linear, and the portfolio problems include execution costs and constraints. We also evaluate the nonlinear forecast

$$\tilde{\mu}_{t,h} = \frac{m_h(a_t)}{1 + m_h(a_t)^\top \Sigma_d^{-1}m_h(a_t)}, \quad m_h(a_t) = B_{\text{CE},h}a_t, \quad (1)$$

called *scaled CE* (SCE) in the tables. As in the earlier experiment, we use $V_h(a_t) \approx \text{Cov}(r) = \Sigma_d$, in daily return units. The unconstrained Markowitz result does not guarantee optimality with costs, constraints, or coupled stages.

Section 5 of [3] explores fast/slow signal weights without specifying a statistical calibration rule. In our earlier alpha-decay net-return study, a grid-selected policy outperformed CE. We include those portfolio problems and re-evaluate the frozen grid winners alongside the gradient fits.

2 Portfolio problems

At date t , observe $a_t \in \mathbb{R}^k$ and previous executed positions $w_0 \in \mathbb{R}^n$. For horizon h , the return input is $\mu_{t,h} = B_h a_t$, with $B_h \in \mathbb{R}^{n \times k}$. Given $\mathcal{B} = (B_1, \dots, B_H)$, solve

$$\begin{aligned} & \underset{w_1, \dots, w_H}{\text{maximize}} && \sum_{h=1}^H \left[(B_h a_t)^\top w_h - C(q_h) - \frac{\gamma}{2} w_h^\top \Sigma_d w_h \right] \\ & \text{subject to} && (w_1, \dots, w_H) \in \mathcal{W}(w_0), \quad q_h = w_h - w_{h-1}. \end{aligned} \quad (2)$$

The variables are planned positions $w_h \in \mathbb{R}^n$; the convex set $\mathcal{W}$ imposes the constraints below at every stage. The risk model uses the unconditional covariance $\Sigma_d = \Sigma_{\text{ann}}/A$, where A is dates per year. All cases except risk aversion have $\gamma = 0$ and $w_h^\top \Sigma_{\text{ann}} w_h \leq .10^2$. Risk-aversion cases use $\gamma = 5$ and no hard risk limit. The following table specifies the six synthetic cases; the stock portfolios are defined in section 4.

Problem	n	H	Execution cost $C(q)$	Extra constraints
Risk only	2	1	0	None
Risk + costs	2	1	$C_2(q) = .0002\ q\ _1 + .0015\ q\ _2^2$	None
Risk + costs + limits	2	1	$C_2(q)$	$ \mathbf{1}^\top w_h \leq .40, \ q_h\ _1 \leq .25,$ $\ w_h\ _1 \leq 1.5$
Risk aversion + costs	2	1	$C_2(q)$	None
Alpha decay: impact	10	2	$.001 \sum_i q_i ^{3/2}$	$\mathbf{1}^\top w_h = 0$
Alpha decay: turnover	10	2	$.001\ q\ _1$	$\mathbf{1}^\top w_h = 0, \ q_h\ _1 \leq .10,$ $\ w_h\ _1 \leq 2$

Execute only w_1 , record net excess return $g_t = w_1^\top r_{t+1} - C(w_1 - w_0)$, and replan at the next date. Positions and costs are fractions of fixed reference capital; gains accumulate in cash without compounding or passive position drift. The same assumed execution losses enter optimization and realized returns. The risk penalty is a preference, not a cash expense.

3 Alphas and conditional expectations

The six synthetic cases have centered, jointly Gaussian returns and alphas, initialized in stationarity. With $\Omega = \text{Cov}(a_t) \succ 0$, vector conditioning gives

$$B_{\text{CE},h} = \text{Cov}(r_{t+h}, a_t) \Omega^{-1}, \quad E[r_{t+h} | a_t] = B_{\text{CE},h} a_t. \quad (3)$$

Every horizon conditions on the same current raw alpha vector.

Two assets. The four single-period problems use $a_t = (a_{f,1,t}, a_{f,2,t}, a_{s,1,t}, a_{s,2,t}) \in \mathbb{R}^4$ and $r_{t+1} \in \mathbb{R}^2$. Specify their joint Gaussian distribution directly:

$$\begin{pmatrix} r_{t+1} \\ a_t \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} \Sigma_d & C_{ra} \\ C_{ra}^\top & \Omega \end{pmatrix}\right), \quad C_{ra} = 10^{-6} \begin{pmatrix} .672 & .056 & .422 & .330 \\ .120 & .764 & .280 & .360 \end{pmatrix}.$$

The sampler draws the return and alpha components of the joint Gaussian state in that order. Gaussian conditioning then gives

$$B_{\text{CE},1} = C_{ra} \Omega^{-1} = \begin{pmatrix} .18 & -.04 & .35 & .12 \\ -.03 & .20 & .10 & .30 \end{pmatrix}.$$

The present comparisons are refitted and evaluated with 10 new seeds. Joint Gaussianity itself implies a linear conditional mean, regardless of the sampling factorization. Each fast alpha has standard deviation .002; the two assets' fast alphas have correlation .30. Slow alphas have standard deviation .001 and correlation .60. Fast and slow blocks are independent, so

$$\Omega = \begin{pmatrix} \Omega_f & 0 \\ 0 & \Omega_s \end{pmatrix}, \quad \Omega_f = .002^2 \begin{pmatrix} 1 & .30 \\ .30 & 1 \end{pmatrix}, \quad \Omega_s = .001^2 \begin{pmatrix} 1 & .60 \\ .60 & 1 \end{pmatrix}.$$

For temporal dependence, $a_{t+1} = \Phi a_t + u_{t+1}$ with $\Phi = \text{diag}(.60, .60, .97, .97)$. Independence of u_{t+1} and a_t implies $\text{Cov}(a_{t+1}) = \Phi \Omega \Phi^\top + \text{Cov}(u_{t+1})$. To retain covariance Ω at every date, choose

$$\text{Cov}(u_{t+1}) = \Omega - \Phi \Omega \Phi^\top = \text{diag}((1 - .60^2)\Omega_f, (1 - .97^2)\Omega_s).$$

This replaces the variance lost through autoregressive decay. Annual return volatilities are 18% and 24%, with correlation .35 and $A = 252$. Each fit learns all eight entries of B_1 .

Ten-asset alpha decay. The two-stage problems retain the earlier note's assumptions: $A = 261$, annual asset volatility 20%, pairwise return correlation .50, fast/slow persistence $p = (2/3, 260/261)$ and noise ratios $\nu = (23.8915392694, 5.7882095765)$. Here $a_t = (a_{f,t}^\top, a_{s,t}^\top)^\top \in \mathbb{R}^{20}$ and $r_{t+h} \in \mathbb{R}^{10}$. Define

$$Q_{jk} = \frac{(1 - p_j)(1 - p_k)}{1 - p_j p_k} (1 + \mathbf{1}_{j=k} \nu_j^2), \quad \Omega = Q \otimes \Sigma_d, \quad c_{h,j} = (1 - p_j) p_j^{h-1}.$$

Equation (3) gives $B_{\text{CE},h} = (c_h^\top Q^{-1}) \otimes I_{10}$. Both fits use $B_h = (K_{hf}, K_{hs}) \otimes I_{10}$ with four independently learned entries of $K \in \mathbb{R}^{2 \times 2}$, initialized at

$$K_{\text{CE}} = \begin{pmatrix} .0029128253 & .0576870695 \\ .0019412497 & .0575211879 \end{pmatrix}.$$

Rows index horizons; columns index fast and slow signals. Cross-asset covariance is included in conditioning and cancels because the covariance blocks share Σ_d . Alphas remain raw and forecast scales are unrestricted.

4 Ten real stocks with artificial alphas

Use adjusted daily prices for AAPL, MSFT, AMZN, GOOGL, META, JPM, XOM, JNJ, WMT and PG from Yahoo Finance [6], subtracting the daily risk-free return from the French data library [7]. The fixed stock list is an ex-post selection, not a historical reconstruction of the S&P 500 universe. Returns are historical and need not be Gaussian.

Let $R_t = (r_{t+1}^\top, r_{t+2}^\top)^\top \in \mathbb{R}^{20}$, with each r_{t+h} a one-day excess return. The current raw alpha vector is

$$a_t = \kappa R_t + \xi_t, \quad \xi_t \sim \mathcal{N}(0, \kappa(1 - \kappa)D), \quad \kappa = \rho^2, \quad \rho = .05,$$

where $D = \text{diag}(\text{Var}_{\text{train}}(R_t))$ and noise is independent across dates and seeds and independent of returns. For each stock and horizon, this gives Pearson $\text{Corr}(a_{h,i,t}, r_{i,t+h}) = \rho$ under the empirical training return distribution. We use this forecast–outcome correlation as IC [5]. The two alpha blocks predict the first and second horizons; neither is standardized. Their noise scales remain fixed after training, so realized IC can change with volatility and sampling variation. These artificial alphas deliberately contain noisy future returns: this is a controlled signal-quality experiment, not an implementable forecasting strategy.

Without Gaussianity, the exact conditional expectation need not be linear. We use the notation $B_{\text{CE},h}$ for the *best linear predictor* in this experiment. Let $M = E_{\text{train}}[R_t R_t^\top]$ and $U = \kappa^2 M + \kappa(1 - \kappa)D$. Integrating out signal noise gives

$$\begin{pmatrix} B_{\text{CE},1} \\ B_{\text{CE},2} \end{pmatrix} = \kappa M U^{-1} = \underset{B \in \mathbb{R}^{20 \times 20}}{\text{argmin}} E_{\text{train},\xi} \|R_t - B a_t\|_2^2. \quad (4)$$

This is regression through the origin; M and U are second moments, not centered covariances. Cross-asset and cross-horizon information enters both maps. All 400 entries are then free in either learning objective.

All three stock problems have $H = 2$, $A = 252$, $C(q) = .0002\|q\|_1 + .0015\|q\|_2^2$, and, at both stages,

$$\mathbf{1}^\top w_h = 0, \quad \|w_h\|_1 \leq 2, \quad |w_{h,i}| \leq .25, \quad |F^\top w_h| \leq (.10, .20).$$

The columns of F are stock betas to SPY and a fixed contrast equal to $+1$ for the first five stocks and -1 for the others. Betas and the risk model use 2013–2017 data; $\Sigma_d = .9S + .1 \text{diag}(S)$ for sample covariance S . The *risk + costs* case has a 10% annual risk cap; *turnover* adds $\|q_h\|_1 \leq .10$; *risk aversion* uses $\gamma = 5$ instead of the risk cap, without the turnover constraint.

5 Fitting through the optimizer

For the Sharpe-learning comparison, use $S = 10$ training paths and compute the mean $\bar{g}_s(\mathcal{B})$ and sample variance $v_s(\mathcal{B})$ of realized net returns over its time horizon. Fit all horizon maps jointly by minimizing

$$L_\lambda(\mathcal{B}) = -\frac{1}{S} \sum_{s=1}^S \frac{\sqrt{A} \bar{g}_s(\mathcal{B})}{\sqrt{v_s(\mathcal{B}) + 10^{-12}}} + \lambda \sum_{h=1}^H \|B_h - B_{\text{CE},h}\|_F^2, \quad (5)$$

starting from $B_h^{(0)} = B_{\text{CE},h}$ for every h . For the shared alpha-decay maps the penalty equals $10\lambda\|K - K_{\text{CE}}\|_F^2$. CVXPY Layers [2] differentiates the solution of (2) with respect to each $B_h a_t$. Autograd carries these derivatives through previous holdings, execution costs, and the realized mean and variance. The penalty adds $2\lambda(B_h - B_{\text{CE},h})$ to each horizon’s gradient.

For the synthetic cases, use 10 independent 504-date training histories and 10 separate 1008-date tuning histories. For each $\lambda \in \{0, .1, 1, 10\}$, take 40 Adam steps from CE. Retain the minimum training-loss iterate for each λ , then choose λ using tuning Sharpe. Freeze the chosen maps before evaluating them on 10 fresh, paired 2610-date test histories. Seed labels are 66000–66009, with independent innovations across the three splits.

For stocks, use 2018 for fitting (250 decisions), 2019–2020 for tuning (504 decisions), and 2021–2025 for testing (1255 decisions). Use 10 independent noise seeds (56000–56009) per split on the same historical returns, 20 Adam steps for each $\lambda \in \{0, .1, 1, 10\}$. Start exactly at (4), apply the same iterate and lambda selection rule. Appendix A replaces the gradient objective with quadratic utility and tunes (η, λ) by achieved tuning Sharpe. Freeze all selected maps for both objectives before test evaluation.

6 Results

The synthetic table reports averages of pathwise annualized Sharpe ratios and paired 95% Student- t intervals for the test difference. Intervals condition on the chosen maps and exclude retraining and multiple-comparison uncertainty. In this section, Fit means the map trained on regularized sample

Sharpe. Every training and test number below is evaluated on the same 10 paths as the other methods in its row.

Problem	Training Sharpe			Test Sharpe			95% interval: fit – CE
	CE	SCE	Fit	CE	SCE	Fit	
Risk only	0.818	0.818	0.851	1.088	1.088	1.061	$[-0.065, +0.011]$
Risk + costs	0.470	0.469	0.723	0.626	0.625	0.596	$[-0.134, +0.074]$
Risk + costs + limits	0.346	0.344	0.444	0.566	0.565	0.626	$[-0.011, +0.130]$
Risk aversion + costs	0.436	0.437	0.461	0.724	0.724	0.722	$[-0.015, +0.010]$
Alpha decay: impact	0.930	0.930	1.012	1.037	1.036	1.105	$[-0.022, +0.157]$
Alpha decay: turnover	0.530	0.533	0.730	0.466	0.458	0.715	$[+0.024, +0.475]$

Positive paired intervals occur for Alpha decay: turnover. The outer loss is nonconvex; gradient fitting provides no certificate of global optimality.

For alpha decay, all fitted maps use the same four-coefficient family as the old grid. Its frozen winners multiply K_{CE} elementwise by $\begin{pmatrix} 1.5 & 1.5 \\ .5 & 1.5 \end{pmatrix}$ for impact and $\begin{pmatrix} 3 & 5 \\ -1 & 5 \end{pmatrix}$ for turnover.

Problem	Test Sharpe				Annual net return			
	CE	SCE	Fit	Grid	CE	SCE	Fit	Grid
Alpha decay: impact	1.037	1.036	1.105	1.093	10.35%	10.34%	11.02%	10.90%
Alpha decay: turnover	0.466	0.458	0.715	0.781	4.54%	4.46%	7.13%	7.80%

Annual net return is arithmetic P&L on reference capital.

Historical stocks. The table compares the linear CE benchmark, scaled CE, and the fitted maps. Its intervals vary only synthetic signal noise: all 10 seeds share the same historical market path. They do not measure uncertainty across alternative market histories.

Problem	Training Sharpe			Test Sharpe			95% interval: fit – CE
	CE	SCE	Fit	CE	SCE	Fit	
Stocks: risk + costs	0.158	0.137	0.597	0.802	0.825	0.853	$[+0.032, +0.070]$
Stocks: turnover	0.264	0.264	2.111	0.491	0.479	0.656	$[-0.201, +0.532]$
Stocks: risk aversion	0.263	0.239	0.667	0.910	0.940	0.949	$[-0.014, +0.091]$

Positive paired intervals occur for Stocks: risk + costs.

Problem	Annual net return			Annual cost		
	CE	SCE	Fit	CE	SCE	Fit
Stocks: risk + costs	8.19%	8.36%	8.71%	6.35%	5.93%	6.27%
Stocks: turnover	4.30%	4.22%	5.85%	0.67%	0.66%	0.67%
Stocks: risk aversion	7.68%	7.70%	7.97%	5.63%	5.21%	5.55%

Mean realized per-stock ICs, averaged across 10 seeds and ordered by horizon, are (0.037, 0.038) for train, (0.059, 0.062) for tune, (0.050, 0.047) for test. Each alpha is paired with its own horizon’s return. For scaled CE on historical stocks, positive paired intervals occur for Stocks: risk aversion.

7 Interpreting the fit through return targets

Restrict the target at horizon h to $\tilde{r}_{t,h} = T_h r_{t+h}$, where $T_h \in \mathbb{R}^{n \times n}$ is fixed. It may mix assets but uses only that horizon’s returns. CE corresponds to $T_h = I_n$. For regression through the origin, the normal equations [4] give

$$\operatorname{argmin}_{M \in \mathbb{R}^{n \times k}} E \|T_h r_{t+h} - M a_t\|_2^2 = T_h E[r_{t+h} a_t^\top] E[a_t a_t^\top]^{-1} = T_h B_{\text{CE},h}.$$

An exact target exists if and only if

$$\hat{B}_h = T_h B_{\text{CE},h}, \quad (6)$$

which requires the rows of $\hat{B}_h$ to lie in the row space of $B_{\text{CE},h}$. To find the closest interpretation, minimize $\|\hat{B}_h - T_h B_{\text{CE},h}\|_F^2$ over T_h . Our CE maps have full row rank, giving the unique solution

$$T_h^* = \hat{B}_h B_{\text{CE},h}^\top (B_{\text{CE},h} B_{\text{CE},h}^\top)^{-1}. \quad (7)$$

We quantify the unrepresented fraction of the fitted coefficients by $e_h = \|\hat{B}_h - T_h^* B_{\text{CE},h}\|_F / \|\hat{B}_h\|_F$. An exact representation requires $e_h = 0$. These statements follow from the normal equations; they are not additional fitted portfolio models.

Two assets. The table gives T_1^* and its mismatch for the Sharpe-trained maps. For example, its first row maps the first asset's target to $T_{11}^* r_{1,t+1} + T_{12}^* r_{2,t+1}$.

Problem	T_{11}^*	T_{12}^*	T_{21}^*	T_{22}^*	e_1
Risk only	1.009	-0.081	-0.002	1.003	5.21%
Risk + costs	1.331	-0.380	0.197	0.402	31.94%
Risk + costs + limits	1.091	-0.064	0.014	0.994	10.28%
Risk aversion + costs	1.017	-0.045	0.043	0.985	4.77%

Ten-asset alpha decay. Write $B_{\text{CE},h} = (c_{hf} I_{10}, c_{hs} I_{10})$ and $\hat{B}_h = (\hat{k}_{hf} I_{10}, \hat{k}_{hs} I_{10})$. An exact target requires

$$c_{hf} T_h = \hat{k}_{hf} I_{10}, \quad c_{hs} T_h = \hat{k}_{hs} I_{10}.$$

Both CE coefficients are nonzero, so T_h must be a scalar multiple of I_{10} and the fast and slow coefficient ratios must agree: $\hat{k}_{hf}/c_{hf} = \hat{k}_{hs}/c_{hs}$. The closest transformation (7) is $T_h^* = s_h I_{10}$.

Problem	h	Fast ratio	Slow ratio	s_h	e_h
Alpha decay: impact	1	0.929	2.347	2.343	3.05%
Alpha decay: impact	2	1.097	2.348	2.347	1.80%
Alpha decay: turnover	1	1.507	2.784	2.781	2.31%
Alpha decay: turnover	2	1.745	2.784	2.783	1.26%

Unequal fast and slow ratios cannot be produced by a transformation of that horizon's returns alone. The reported Sharpes use the unconstrained input maps. Requiring an exact target interpretation during fitting would restrict the policy family to $B_h = T_h B_{\text{CE},h}$.

Historical stocks. The same-horizon return transformations may mix the ten stocks. Mean own-return weight is $\text{tr}(T_h^*)/10$ and maximum cross-stock weight is $\max_{i \neq j} |(T_h^*)_{ij}|$; their benchmark values are 1 and 0. The mismatch e_h tests whether these return-only targets can represent the fit.

Problem	h	Mean own	Max cross	e_h
Stocks: risk + costs	1	0.993	0.081	1.4%
Stocks: risk + costs	2	0.998	0.045	1.2%
Stocks: turnover	1	1.011	0.299	8.1%
Stocks: turnover	2	1.033	0.355	7.1%
Stocks: risk aversion	1	0.995	0.057	1.4%
Stocks: risk aversion	2	0.995	0.046	1.2%

A Learning with regularized quadratic utility

For learning $\mathcal{B}$, we replace sample Sharpe with regularized quadratic utility of realized net returns and keep Sharpe as the evaluation metric. This is because Sharpe estimators tend to have high standard errors, particularly over short histories [5, chapter 17]. The quadratic-utility loss avoids division by estimated volatility. Let $w_t(\mathcal{B})$ be the executed position, and define

$$g_t(\mathcal{B}) = w_t(\mathcal{B})^\top r_{t+1} - C(w_t(\mathcal{B}) - w_{t-1}(\mathcal{B})).$$

Using averages over all training dates and all ten seeds, minimize

$$L_{\eta,\lambda}(\mathcal{B}) = -\widehat{E}[g_t(\mathcal{B})] + \frac{\eta}{2}\widehat{E}[g_t(\mathcal{B})^2] + \lambda \sum_{h=1}^H \|B_h - B_{\text{CE},h}\|_F^2. \quad (8)$$

The quadratic term is a *second moment*, not a centered variance. For fixed (η, λ) , the gradient is

$$\nabla_{B_h} L_{\eta,\lambda} = \widehat{E}[(\eta g_t - 1)\nabla_{B_h} g_t] + 2\lambda(B_h - B_{\text{CE},h}).$$

CVXPY Layers and autograd propagate this gradient through all planned stages and carried positions. Each candidate starts exactly at $B_{\text{CE},h}$.

Connection to Sharpe. The following calculation explains the connection without a distributional assumption. Write $\mu = E[g] > 0$, $q = E[g^2] > \mu^2$, and let $s = \mu/\sqrt{q - \mu^2}$ be the unannualized Sharpe ratio. For a return stream whose exposure can be scaled freely by $c \geq 0$,

$$\max_{c \geq 0} \left\{ c\mu - \frac{\eta}{2}c^2q \right\} = \frac{\mu^2}{2\eta q} = \frac{1}{2\eta} \frac{s^2}{1 + s^2}, \quad c^* = \frac{\mu}{\eta q}.$$

The optimized utility is increasing in positive Sharpe. This is our direct derivation; free scaling need not hold for the constrained portfolios with nonlinear execution costs. Consequently, utility is a related learning criterion, not an exact reformulation of every portfolio Sharpe problem. The outer problem remains nonconvex and its second moments still have sampling error.

Choosing η and λ . From the CE training P&L alone, define

$$\eta_0 = \frac{\widehat{E}[g_{\text{CE}}]}{\widehat{E}[g_{\text{CE}}^2]}, \quad u_0 = \widehat{E}[g_{\text{CE}}] - \frac{\eta_0}{2}\widehat{E}[g_{\text{CE}}^2] > 0.$$

The reference η_0 makes $c = 1$ optimal along the CE scaling direction. Search $\eta \in \{.25, 1, 4\}\eta_0$ and $\lambda \in \{0, .1, 1, 10\}u_0$, with the same ten paths, gradient steps and full-matrix penalty as the Sharpe fits. For each pair, retain its minimum regularized *training-utility loss* iterate. Select the pair with the largest achieved *tuning Sharpe*, then freeze it before testing. Thus Sharpe selects hyperparameters but is never differentiated in utility learning. Dividing the entire loss by the fixed positive u_0 only improves numerical scaling and does not change its minimizer.

Training and test performance. Here QU denotes the utility-trained map. CE and SCE repeat the same baselines as the main tables; no evaluation changes the inner cost or risk model.

Problem	Training Sharpe			Test Sharpe			95% interval: QU – CE
	CE	SCE	QU	CE	SCE	QU	
Risk only	0.818	0.818	0.860	1.088	1.088	1.058	$[-0.074, +0.014]$
Risk + costs	0.470	0.469	0.594	0.626	0.625	0.613	$[-0.080, +0.055]$
Risk + costs + limits	0.346	0.344	0.678	0.566	0.565	0.657	$[-0.049, +0.231]$
Risk aversion + costs	0.436	0.437	0.519	0.724	0.724	0.656	$[-0.111, -0.025]$
Alpha decay: impact	0.930	0.930	1.012	1.037	1.036	1.107	$[-0.017, +0.157]$
Alpha decay: turnover	0.530	0.533	0.705	0.466	0.458	0.707	$[+0.032, +0.450]$
Stocks: risk + costs	0.158	0.137	0.823	0.802	0.825	0.866	$[+0.010, +0.118]$
Stocks: turnover	0.264	0.264	1.970	0.491	0.479	0.741	$[-0.063, +0.563]$
Stocks: risk aversion	0.263	0.239	0.936	0.910	0.940	0.967	$[-0.030, +0.144]$

For quadratic-utility learning, positive paired intervals occur for Alpha decay: turnover, Stocks: risk + costs. Negative intervals occur for Risk aversion + costs. The intervals use ten paired seed differences. For historical stocks they reflect artificial-signal noise on one observed return history. The broader utility search has twelve candidates per problem versus four for Sharpe learning; this comparison does not isolate the loss from search budget.

Conclusion

Learning alpha maps through convex portfolio optimization can improve out-of-sample Sharpe over CE in some problems. Both learning objectives yield positive paired intervals for alpha-decay turnover and the historical-stock risk-and-costs case. These gains do not establish that performance is preserved in other problems. For the Sharpe-trained maps, no paired interval is wholly negative, but several point estimates are lower than CE and their intervals allow for losses.

Quadratic-utility learning reduces test Sharpe in the two-asset risk-aversion case from 0.724 to 0.656, with a paired 95% interval for the change of $[-0.111, -0.025]$. This is direct evidence of underperformance in that experiment. Initializing at CE and regularizing towards it does not guarantee performance on new data. Gains from fitting depend on the portfolio problem, and losses remain possible. Overall, conditional expectation performs well and remains a strong baseline in these experiments.

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